

Installing Ruby:

1. Download Ruby Installer:

- **For Windows:** Go to the RubyInstaller website (<https://rubyinstaller.org/>) and download the installer.
- **For macOS:** Use Homebrew by running `brew install ruby`.
- **For Linux:** Use your distribution's package manager, e.g., `sudo apt-get install ruby-full` for Ubuntu.

2. Install Ruby:

- Follow the installation instructions specific to your operating system.
- Verify the installation by opening a terminal and running `ruby -v`. This should display the installed Ruby version.

Setting Up a Development Environment:

1. Choose a Text Editor/IDE:

- Popular choices include Visual Studio Code, Sublime Text, Atom, and RubyMine.
- **Visual Studio Code:** Download from Visual Studio Code (<https://code.visualstudio.com/>).
- **RubyMine:** A commercial IDE from JetBrains, specifically designed for Ruby development.

2. Install Ruby Extensions:

- In your chosen text editor or IDE, install Ruby extensions or plugins to enhance Ruby development. For example, in Visual Studio Code, you can install the “Ruby” extension from the marketplace.

Basic Ruby Syntax

Understanding the basic syntax of Ruby is essential for writing your first program. Here are some fundamental concepts and constructs:

Variables and Data Types:

Ruby supports various data types, including integers, floats, strings, arrays, hashes, and symbols. Variables do not require explicit declaration and can change type dynamically.

```
# Integer
age = 25

# Float
height = 5.9

# String
name = "Alice"

# Boolean
is_student = true
```



```
# Array
numbers = [1, 2, 3, 4, 5]

# Hash
person = { name: "Alice", age: 25 }

# Symbol
status = :active

puts "Name: #{name}, Age: #{age}, Height:
#{height}, Is Student: #{is_student}"
```

Comments:

Comments in Ruby can be single-line, starting with #, or multi-line, enclosed in `=begin` and `=end`.

```
# This is a single-line comment

=begin
This is a
multi-line comment
=end
```

Control Structures:

Ruby supports control structures like conditionals and loops.

```
# Conditional
age = 18
if age >= 18
  puts "You are an adult."
else
  puts "You are a minor."
end

# Loop
5.times do |i|
  puts "Iteration: #{i}"
end
```

Methods:

Methods in Ruby are defined using the `def` keyword and can return a value.